

Lecture 4: Properties of OLS

Economics 326 — Introduction to Econometrics II

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Properties of Estimators

1. Random
2. Mean
3. Variance
4. Distribution

OLS Estimators as Random Variables

- The model

$$Y_i = \alpha + \beta X_i + U_i,$$
$$E[U_i | X_1, \dots, X_n] = 0.$$

Conditioning on $X_1, \dots, X_n$ allows us to treat all the X_i 's as fixed, but Y_i is still random.

- To save on writing, we will use the notation

$$E[\cdot | \mathbf{X}] = E[\cdot | X_1, \dots, X_n].$$

That is, $\mathbf{X} = (X_1, \dots, X_n)$.

- The estimators

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \text{ and } \hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$$

are random because they are functions of the random Y_i 's even after conditioning on the X_i 's.

Linearity of Estimators

- Since

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2},$$

we can write $\hat{\beta} = \sum_{i=1}^n w_i Y_i$, where

$$w_i = \frac{X_i - \bar{X}}{\sum_{l=1}^n (X_l - \bar{X})^2}.$$

After conditioning on X 's, w_i 's are not random.

- For $\hat{\alpha}$,

$$\begin{aligned}
\hat{\alpha} &= \bar{Y} - \hat{\beta} \bar{X} \\
&= \frac{1}{n} \sum_{i=1}^n Y_i - \left(\sum_{i=1}^n w_i Y_i \right) \bar{X} \\
&= \sum_{i=1}^n \left(\frac{1}{n} - \bar{X} w_i \right) Y_i \\
&= \sum_{i=1}^n \left(\frac{1}{n} - \bar{X} \frac{X_i - \bar{X}}{\sum_{l=1}^n (X_l - \bar{X})^2} \right) Y_i.
\end{aligned}$$

Unbiasedness

Definition and Claim

- $\hat{\beta}$ is called an unbiased estimator if $E[\hat{\beta}] = \beta$.
- Claim: Suppose that
 - $Y_i = \alpha + \beta X_i + U_i$,
 - $E[U_i | \mathbf{X}] = 0$.
 - Then

$$E[\hat{\beta}] = \beta.$$

Proof Step 1: Decomposition into signal and noise

$$\begin{aligned}
\hat{\beta} &= \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \\
&= \frac{\sum_{i=1}^n (X_i - \bar{X}) (\alpha + \beta X_i + U_i)}{\sum_{i=1}^n (X_i - \bar{X})^2} \\
&= \alpha \frac{\sum_{i=1}^n (X_i - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2} + \beta \frac{\sum_{i=1}^n (X_i - \bar{X}) X_i}{\sum_{i=1}^n (X_i - \bar{X})^2} + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \\
&= \alpha \frac{0}{\sum_{i=1}^n (X_i - \bar{X})^2} + \beta \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2} + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \\
\hat{\beta} &= \underbrace{\beta}_{\text{signal}} + \underbrace{\frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}}_{\text{noise}}
\end{aligned}$$

Proof Step 2: Conditioning on Regressors

- Once we condition on $\mathbf{X}$, all the X_i 's in

$$\hat{\beta} = \beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

can be treated as fixed.

- Thus,

$$\begin{aligned} E[\hat{\beta} | \mathbf{X}] &= E\left[\beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \mid \mathbf{X}\right] \\ &= \beta + E\left[\frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \mid \mathbf{X}\right] \\ &= \beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) E[U_i | \mathbf{X}]}{\sum_{i=1}^n (X_i - \bar{X})^2}. \end{aligned}$$

Proof Step 3

- Thus, with $E[U_i | \mathbf{X}] = 0$, we have

$$\begin{aligned} E[\hat{\beta} | \mathbf{X}] &= \beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) E[U_i | \mathbf{X}]}{\sum_{i=1}^n (X_i - \bar{X})^2} \\ &= \beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) \cdot 0}{\sum_{i=1}^n (X_i - \bar{X})^2} = \beta. \end{aligned}$$

- By the LIE, $E[\hat{\beta}] = E[E[\hat{\beta} | \mathbf{X}]] = E[\beta] = \beta$.

Strong Exogeneity of Regressors

- $\mathbf{X} = (X_1, \dots, X_n)$ are strongly exogenous if $E[U_i | \mathbf{X}] = 0$.
- Alternatively, we can assume that $E[U_i | X_i] = 0$ and all observations are independent:

$$\begin{aligned} E[U_1 | \mathbf{X}] &= E[U_1 | X_1], \\ E[U_2 | \mathbf{X}] &= E[U_2 | X_2] \text{ etc.} \end{aligned}$$

- The OLS estimator is in general biased if the strong exogeneity assumption is violated.

Variance of the Slope Estimator

Variance Formula and Homoskedasticity

- If $Y_i = \alpha + \beta X_i + U_i$, $E[U_i | \mathbf{X}] = 0$, and

$$E[U_i^2 | \mathbf{X}] = \sigma^2 = \text{constant},$$

and for $i \neq j$

$$E[U_i U_j | \mathbf{X}] = 0,$$

then

$$\text{Var}(\hat{\beta} | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- The assumption $E[U_i^2 | \mathbf{X}] = \sigma^2 = \text{constant}$ is called (conditional) homoskedasticity.
- The assumption $E[U_i U_j | \mathbf{X}] = 0$ for $i \neq j$ can be replaced by the assumption that the observations are independent.

Determinants of Variance

$$\text{Var}(\hat{\beta} | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- The variance of $\hat{\beta}$ is positively related to the variance of the errors $\sigma^2 = \text{Var}(U_i)$.
- The variance of $\hat{\beta}$ is smaller when the X_i 's are more dispersed.

Derivation of Variance

- We have $\hat{\beta} = \beta + \frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}$ and $E[\hat{\beta} | \mathbf{X}] = \beta$.

$$\begin{aligned} \text{Var}(\hat{\beta} | \mathbf{X}) &= E\left[\left(\hat{\beta} - E[\hat{\beta} | \mathbf{X}]\right)^2 | \mathbf{X}\right] \\ &= E\left[\left(\frac{\sum_{i=1}^n (X_i - \bar{X}) U_i}{\sum_{i=1}^n (X_i - \bar{X})^2}\right)^2 | \mathbf{X}\right] \\ &= \left(\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2}\right)^2 E\left[\left(\sum_{i=1}^n (X_i - \bar{X}) U_i\right)^2 | \mathbf{X}\right]. \end{aligned}$$

- Expanding the square,

$$\begin{aligned} \left(\sum_{i=1}^n (X_i - \bar{X}) U_i\right)^2 &= \sum_{i=1}^n \sum_{j=1}^n (X_i - \bar{X})(X_j - \bar{X}) U_i U_j \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 U_i^2 \\ &\quad + \sum_{i=1}^n \sum_{j \neq i} (X_i - \bar{X})(X_j - \bar{X}) U_i U_j. \end{aligned}$$

- Since $E[U_i U_j | \mathbf{X}] = 0$ for $i \neq j$,

$$\begin{aligned} E\left[\left(\sum_{i=1}^n (X_i - \bar{X}) U_i\right)^2 | \mathbf{X}\right] &= \sum_{i=1}^n (X_i - \bar{X})^2 E[U_i^2 | \mathbf{X}] + 0 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 \sigma^2. \end{aligned}$$

- We have

$$\begin{aligned} \text{Var}(\hat{\beta} | \mathbf{X}) &= \left(\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2}\right)^2 E\left[\left(\sum_{i=1}^n (X_i - \bar{X}) U_i\right)^2 | \mathbf{X}\right], \\ E\left[\left(\sum_{i=1}^n (X_i - \bar{X}) U_i\right)^2 | \mathbf{X}\right] &= \sigma^2 \sum_{i=1}^n (X_i - \bar{X})^2, \end{aligned}$$

and therefore,

$$\begin{aligned} \text{Var}(\hat{\beta} | \mathbf{X}) &= \left(\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2}\right)^2 \sigma^2 \sum_{i=1}^n (X_i - \bar{X})^2 \\ &= \left(\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2}\right) \sigma^2. \end{aligned}$$

Distribution of the Slope Estimator

Normality of the OLS Estimator

- Assume that U_i 's are jointly normally distributed conditional on X 's.
- Then $Y_i = \alpha + \beta X_i + U_i$ are also jointly normally distributed.

- Since $\hat{\beta} = \sum_{i=1}^n w_i Y_i$, where $w_i = \frac{X_i - \bar{X}}{\sum_{i=1}^n (X_i - \bar{X})^2}$ depend only on the X_i 's, $\hat{\beta}$ is also normally distributed conditional on the X_i 's.
- Conditional on $\mathbf{X}$

$$\begin{aligned}\hat{\beta} \mid \mathbf{X} &\sim N\left(\mathbb{E}[\hat{\beta} \mid \mathbf{X}], \text{Var}(\hat{\beta} \mid \mathbf{X})\right) \\ &\sim N\left(\beta, \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}\right).\end{aligned}$$